DEMANDIQ · PEAK-SEASON PLANNING BRIEF

Plan the outerwear peak on demand, not on last year's sales

Premium outerwear demand planning case study · Abhijit Mishra · simulated data, October 2026

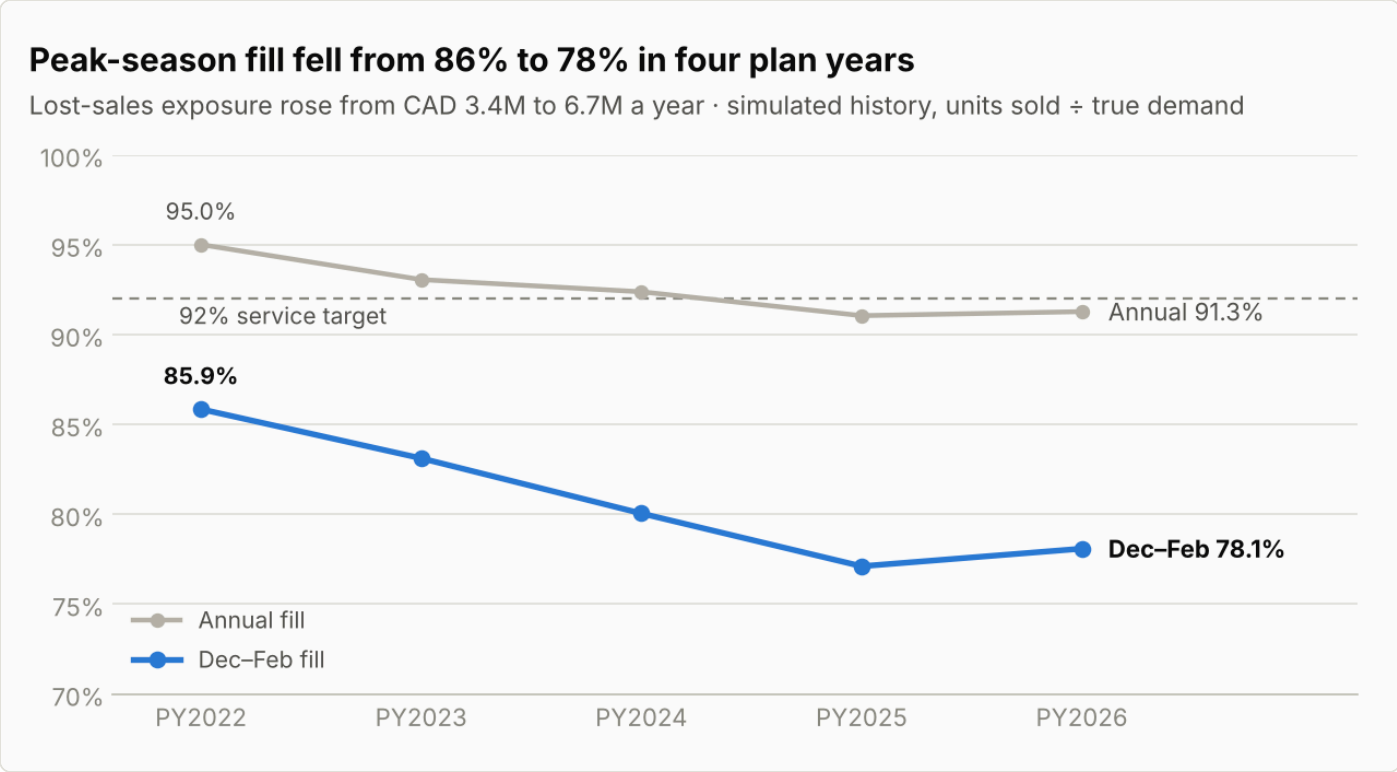
Peak-season fill fell from 86% to 78% in four plan years because every buy starts from last year's sales, and stockouts suppress exactly the peak weeks. Planning the next peak on reconstructed demand means an indicative **10,339 more units (+21%)**, and the order window closes on **Jun 29–Jul 13, 2026**.

78.1%
Dec–Feb fill in PY2026, down from 85.9% *(truth)*

CAD 6.65M
lost revenue, 9.1% of net revenue *(truth, upper bound)*

+10,339 units
indicative next-peak change: 4,574 moved, 5,613 new *(observable)*

Jun 29–Jul 13
last order dates on the 22-week lane for arrival before the first stockout



Why it happens

- **The buy plans the business down.** Each buy starts from last year's sales, 9.0% below last year's reconstructed demand. In Dec–Feb, reconstructed demand runs 21% above sales. observable
- **Most of the loss is timing.** 59% of lost units had stock arriving within the plan year, but after the stockouts. truth
- **Chase is aimed at the wrong signal.** It fires in mid-September in 75–77% of cells whatever their eventual loss, and 5,403 units of chase capacity went unused. observable
- **The current 13-week plan looks at the wrong season.** Its summer window holds 1.1% of lost demand. There, weather and stockout correction change none of the 9 planner actions; the supply assumption changes 6. observable

Evidence labels: **observable** = sales, in-stock days, receipts, inventory and reconstructed demand, which a planner can see. **Truth** = hidden simulation fields, used only to size the problem and to validate.

What to do

Decision	Quantity	By when · owner
1. Plan the peak on reconstructed demand; move stock before buying	Indicative +10,339: 4,574 moved forward from spring replenishment, 5,613 new, 152 accepted gap (excess guardrail)	Supplier commitment by Jun 29 (6 series) or Jul 13 (3 series) · planner proposes, S&OP approves, supply confirms
2. Size chase from the peak gap, not early shipments	Fallback floor of about 4,355 units within last season's unused chase capacity	Chase order by Sep 21 or Oct 5 · planner with supply planning
3. Rank exceptions by stockout days, by SKU × region × channel	Top 20 of 81 cells hold 60% of lost revenue	Each monthly S&OP · Demand Planner
4. Escalate only shortfalls that survive forecast error	0 escalate, 1 watch, 5 protect, 3 no action (the original list had 3 escalations)	Weekly S&OE
5. Release launch reserve on a 5-pp channel-mix rule	991-unit reserve on an 8,259-unit buy; ECOM lost 159 units while it sat idle	Launch checkpoints W1–W2

Peak scenario by series observable

Series	Gap vs sales	Change	Moved	New	Call
IMH-001 ECOM	+20%	2,355	1,308	1,047	Buy and re-time
APS-001 ECOM	+21%	960	593	367	Buy and re-time
APS-001 Retail	+32%	958	343	615	Buy and re-time
IMH-001 Retail	+18%	1,599	674	925	Buy and re-time
IMH-001 Wholesale	+46%	2,492	927	1,565	Buy and re-time
CTS-001 Retail	+15%	625	161	464	Buy and re-time
APS-001 Wholesale	+27%	690	212	478	Buy and re-time
CTS-001 Wholesale	+11%	341	189	152	Buy and re-time
CTS-001 ECOM	+5%	319	167	0	Re-time only
Total		**10,339**	**4,574**	**5,613**	**152 units accepted as a gap**

Change = (reconstructed − sold) Dec–Feb units × (1 + SKU growth). It is allocated to regions by stockout days. Stock that was unsold at season end and arrived as spring replenishment is moved first; the rest is new units. No new units where over-buy exposure ≥ 50% of estimated lost revenue (a policy assumption).

Can it land in time?

- **New units:** the regular October-drop order date (Apr 27) has passed. New 22-week orders must be placed by Jun 29–Jul 13; supplier capacity is **unconfirmed**.
- **Moved units:** they must ship 17–21 weeks earlier than last season's spring receipts, which needs a supplier order change. Order-change windows are not modelled: **unconfirmed**.
- **Fallback:** chase alone lands about 4,355 units by ordering by Sep 21–Oct 5. New units fit last season's chase headroom in 3 of 8 series.

How sure are we?

- **Direction: high.** Scored against simulation truth, the priority ranking matches 9 of 9 and every call survives recomputation with true demand. truth
- **Size: conservative.** Reconstruction finds 75% of the true Dec–Feb gap. The planner view also asks for more new units than truth suggests (54% vs 41%). truth
- **Limits.** The data is simulated, and the peak throttle and shipment-based chase are designed into the generator; the test is that the diagnosis finds them from observable data. Lost revenue assumes no substitution. Supplier capacity, MOQs and order-change windows are not modelled.

Reproduce every number: **python 04_scripts/step8/step8_peak_diagnosis.py** · github.com/Abhijit2409/demandiq-retail-demand-planning · Portfolio simulation using public weather and synthetic demand, inventory and launch data; no company's internal data. Economic values are planning-exposure proxies, not accounting profit.